

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring-2023 Mid-term Examination
Program: B.Sc. in CSE (Batch – 15th) 1st Year 1st Semester
Course Title: Physics - I
Course Code: PHY 1101

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. What is the value of the permittivity of free space?

- A) $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$
- B) $9 \times 10^9 Nm^2 C^{-2}$
- C) $1.9 \times 10^{-19} C$
- D) $1.6 \times 10^{-19} C$
- E) $9.1 \times 10^{-31} kg$

Cognitive Knowledge

2. Which is equal to the reciprocal of conductivity?

- A) Resistivity
- B) Specific resistance
- C) Conductance
- D) Conductivity
- E) Permittivity

Cognitive Understanding

3. What does a dielectric medium do?

- A) Insulates the conductor
- B) Increases current flow
- C) Produces anti-electric field
- D) Boost up the collision of electrons
- E) Store the charges

Cognitive Knowledge

4. Which term is the equivalent of Ohm (Ω)?

- A) Am^{-1}
- B) $Csec^{-1}$
- C) Fm^{-1}
- D) AV^{-1}
- E) VA^{-1}

Cognitive Understanding

5. How many conductors are required to form a capacitor?

- A) 0
- B) 1
- C) 2
- D) 4
- E) 8

Cognitive Application

6. Which is the correct equation for the energy stored in a capacitor?

- A) $w = \frac{1}{2} CV$
- B) $w = \frac{1}{2} CV^2$
- C) $w = \frac{1}{2} VQ$
- D) $w = \frac{1}{2} CQ^2$
- E) $w = \frac{1}{2} VC^2$

Cognitive Understanding

7. In which Coulomb's law is applicable?

- A) Distributed charges
- B) Moving charges
- C) Point charges
- D) Charged particles of any shape
- E) Charged particles of any size

Cognitive Application

8. Two wires each of radius of cross-section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1:4, the drift velocity of electrons in the two wires will be in the ratio:

- A) 1:2
- B) 2:1
- C) 1:4
- D) 4:1
- E) 1:8

Cognitive Understanding

9. Which one is possible?

- A) $q = 45e$
- B) $q = 35.1e$
- C) $q = 25.5e$
- D) $q = 15.9e$
- E) $q = 5.5e$

Cognitive Knowledge

10. $\int \vec{j} \cdot \vec{da} = ?$

- A) Current flow (I)
- B) Electric potential (V)
- C) Electric field (E)
- D) Resistance (R)
- E) Conduction (G)

Signature of Examiner

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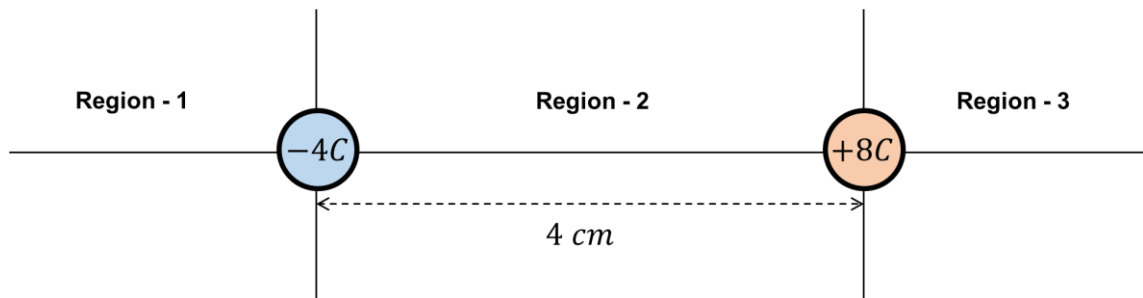
PART-B: SAQ

Time: 1 Hour and 20 Minutes

Total Marks: 4x5=20

INSTRUCTIONS: Answer question no. 01 (one) and any 03 (three) from the rest. Marks are shown by side.

1. See the stem carefully and answer the question that follows:



- a) Define 1Coulomb. [1]
b) Characterize the electric field lines by the diagram for the above stem. [1]
c) Select a region in which the electric field will be zero and give the logical explanation mathematically. [3]
2. a) What is relative permittivity? [1]
b) Why a charged an object is said to be quantized? [1]
c) A copper wire of 3.0mm^2 cross-sectional area carries a current of 5.0A . Find the magnitude of the drift velocity for the electrons in the wire. [3]
3. a) State the Gauss law. [2]
b) Calculate the electric flux for the following shaped object placed in an external field: [3]



4. a) What will be happened if you fill the capacitor with dielectric materials? [1]
b) The parallel plates of an air-filled capacitor are everywhere 1.0mm apart. What must be the plate area if the capacitance is to be 1.0F ? [2]
c) Proof that "the capacitance of a cylindrical capacitor is independent of charge of the capacitor and the potential difference between the plates of the capacitor". [2]
5. a) Define electric field by potential gradient. [1]
b) Express the Ohm's law in terms of current density. [2]
c) Relate the resistivity with the number of electrons per unit volume. [2]